ALP Dark Matter, Cosmological Magnetic Fields and the Direct Collapse Black Hole Formation Scenario

Working out the Paper by Kushwaha & Brandenberger (2026)

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Preface

These notes are a complete and self-contained study companion to the paper *ALP Dark Matter, Cosmological Magnetic Fields and the Direct Collapse Black Hole Formation Scenario* by Ashu Kushwaha and Robert Brandenberger (arXiv:2602.22170, February 2026). They build directly on the preceding notes for Kamali & Brandenberger (arXiv:2601.00443), which treated scalar dark matter coupled to electromagnetism via a gauge-kinetic function $I(\phi)F^2$. The present paper concerns the complementary *pseudoscalar* (axion-like) case $g_{\phi\gamma}\phi F \wedge F$, and addresses an entirely new physical question: can the cosmological magnetic field generated by this mechanism, together with the ambient dark matter fluctuations, produce a sufficient flux of Lyman-Werner (LW) photons to enable Direct Collapse Black Hole (DCBH) formation?

The logical flow is:

1. Review of the context: the puzzle of supermassive black holes at high redshift and the DCBH scenario.
2. Review of the ALP magnetogenesis mechanism from the companion paper [1].
3. The new mechanism: secondary photons produced by the coupling of the cosmological magnetic field B to the dark matter density fluctuations $\delta\phi$.
4. Green's function solution for the induced gauge field.
5. Computation of the secondary photon flux and the LW criterion.
6. Constraints from excess radio photon experiments (ARCADE2, EDGES).
7. The viable parameter window and open questions.

Notation. Primes ($'$) denote derivatives with respect to conformal time η. Dots ($\dot{}$) denote derivatives with respect to cosmic time t. We use natural units $c = \hbar = k_B = 1$ throughout. The reduced Planck mass is $M_{\rm pl} = (8\pi G)^{-1/2} \simeq 2.435 \times 10^{27}\,\text{eV}$. The metric signature is $(+, -, -, -)$ throughout this document (matching the convention in the appendix of Kushwaha & Brandenberger).

1 Physical Motivation

1.1 The Supermassive Black Hole Puzzle

Supermassive black holes (SMBHs) with masses $M_\bullet \gtrsim 10^9\,M_\odot$ are observed as quasars at redshifts $z \gtrsim 6$, implying they existed when the universe was less than 10^9 years old. This is deeply puzzling: a stellar-mass black hole ($M_\bullet \sim 10\,M_\odot$) accreting at the Eddington limit and doubling its mass every $t_{\rm Sal} \simeq 45\,\mathrm{Myr}$ requires ≈ 20 e-folding times to reach $10^9\,M_\odot$, corresponding to $\approx 10^9\,\mathrm{yr}$—barely consistent with observations, and only if accretion was uninterrupted. Any reasonable delay or sub-Eddington episode makes the timeline untenable.

Two broad classes of solutions exist:

- **Primordial black holes**: the seeds were formed in the very early universe (e.g. during radiation domination). These are not the focus of this paper.
- **Late-time seeding**: very massive (10^4–$10^6\,M_\odot$) seed black holes form through the direct collapse of massive gas clouds at $z \sim 10$–15. This is the *Direct Collapse Black Hole* (DCBH) scenario.

1.2 The Direct Collapse Black Hole Scenario

For a collapsing gas cloud to form a DCBH rather than fragmenting into ordinary stars, two conditions must be met:

I. **Sufficient nonlinear seed fluctuation.** There must be a perturbation on the relevant mass scale that has gone nonlinear and provides a potential well for the gas to collapse into.

II. **Prevention of fragmentation.** The infalling gas must not cool below the atomic hydrogen cooling threshold ($T \sim 10^4\,\mathrm{K}$), because once molecular hydrogen (H_2) forms it acts as an efficient coolant, causing the cloud to fragment into many small ($\sim M_\odot$) pieces instead of collapsing monolithically.

The key to condition II is the suppression of H_2 formation. H_2 is dissociated by photons in the *Lyman-Werner* (LW) band, covering photon energies 11.2–13.6 eV (corresponding to momenta $k_{\rm LW} \sim 10\,\mathrm{eV}$ in natural units). These photons are energetic enough to break the H_2 molecule (via photodissociation of the B and C electronic states) but not to ionise hydrogen (13.6 eV).

DCBH Criterion

A sufficient flux of LW photons,

$$J_{\rm LW} \gtrsim J_{\rm LW,crit} \sim 10^3\ J_{21}, \quad J_{21} \equiv 10^{-21}\,{\rm erg\,s^{-1}cm^{-2}Hz^{-1}sr^{-1}}, \tag{1.1}$$

is required to prevent H_2 cooling and enable monolithic collapse of a gas cloud into a DCBH.

The standard astrophysical sources of LW photons (young stars in neighbouring galaxies) are typically insufficient unless a very specific geometric alignment exists. A cosmological source of LW photons would therefore be highly attractive.

1.3 ALP Dark Matter as a Common Source

The key insight of Kushwaha & Brandenberger is that an axion-like particle (ALP) serving as the dark matter can simultaneously:

1. Generate cosmological magnetic fields through a tachyonic instability right after recombination (the mechanism established in [1]).
2. Use that same magnetic field together with the primordial dark matter density fluctuations to produce secondary photons via the coupling $g_{\phi\gamma}\phi F \wedge F$, reaching the Lyman-Werner band.

This is a two-step process. The first step ($\phi \to A_k + A_{-k}$) generates a nearly uniform cosmological magnetic field $B \sim 1\,{\rm G}$ on scale $k_{\rm c}$. The second step ($\delta\phi_k + B \to \delta A_k$) uses the background B as a catalyst to convert dark matter fluctuations into photons.

Note

The process is analogous to inverse Compton scattering in a magnetic field: the oscillating ALP condensate fluctuation $\delta\phi$ acts as the "electron," the background B provides the seed photon, and the output is a propagating electromagnetic wave (secondary photon).

2 Review of ALP Magnetogenesis

This chapter summarises the mechanism of Brandenberger, Fröhlich, and Jiao [1] which produces the background magnetic field B that appears as an input in the main calculation of the present paper.

2.1 The ALP Lagrangian

The dynamics of an axion-like particle (ALP) field ϕ coupled to electromagnetism are described by the Lagrangian

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - g_{\phi\gamma}\,\phi\,F\wedge F, \tag{2.1}$$

where $F\wedge F \equiv F_{\mu\nu}\tilde{F}^{\mu\nu} = -4\boldsymbol{E}\cdot\boldsymbol{B}$ (the pseudoscalar Chern-Simons term), $g_{\phi\gamma}$ is the coupling constant with dimensions of inverse mass, and m is the ALP mass.

The coupling term is odd under parity ($\phi \to -\phi$ for a pseudoscalar, $\boldsymbol{E}\cdot\boldsymbol{B} \to -\boldsymbol{E}\cdot\boldsymbol{B}$), making the product $\phi F\wedge F$ a Lorentz scalar, as required for a consistent Lagrangian.

2.2 Parametrisation of the Parameters

The ALP mass and coupling are written as

$$m \equiv m_{20} \times 10^{-20}\,\mathrm{eV}, \tag{2.2}$$

$$g_{\phi\gamma} \equiv \tilde{g}_{\phi\gamma} \times 10^{-10}\,\mathrm{GeV}^{-1}, \tag{2.3}$$

where m_{20} and $\tilde{g}_{\phi\gamma}$ are dimensionless numbers of order unity. Current constraints require $m_{20} > 10$ (from ultra-faint dwarf galaxies) and $g_{\phi\gamma} < 10^{-9}\,m_{20}\,\mathrm{GeV}^{-1}$ (i.e. $\tilde{g}_{\phi\gamma} < 10\,m_{20}$).

2.3 Coherent Oscillations via Misalignment

If the ALP is produced by a misalignment mechanism, the field starts frozen at some initial displacement $\phi_{\mathrm{init}} \neq 0$ when the Hubble rate $H \gg m$. Once H drops below m, the field begins to oscillate:

$$\phi(\eta) = \phi_0\cos(m\eta), \tag{2.4}$$

where the amplitude ϕ_0 is related to the DM energy density. If ϕ constitutes all of the dark matter at recombination:

$$\frac{1}{2}m^2\phi_0^2 \sim T_{\mathrm{rec}}^4 \quad \Longrightarrow \quad \phi_0 \sim \frac{T_{\mathrm{rec}}^2}{m}. \tag{2.5}$$

The field is coherent over the entire Hubble patch, sidestepping the correlation-length problem that afflicts phase-transition mechanisms.

2.4 Tachyonic Resonance for Gauge Modes

The mode equation for the two circular polarisations $A_k^\pm$ of the gauge field, derived from the Lagrangian (2.1) in Coulomb gauge, takes the form (in conformal time)

$$\ddot{A}_k^\pm + \left(k^2 \pm g_{\phi\gamma}\,\dot{\phi}\,k\right) A_k^\pm = 0. \tag{2.6}$$

For one of the two polarisations (say A_k^-), when $g_{\phi\gamma}|\dot{\phi}|k > k^2$, the effective frequency-squared becomes negative, signalling a *tachyonic instability*: the mode grows exponentially rather than oscillating.

Critical wavenumber
The tachyonic instability occurs for modes with wavenumber $k < k_\mathrm{c}$, where

$$k_\mathrm{c} \simeq g_{\phi\gamma}\, m\, \phi_0\, a(\eta). \tag{2.7}$$

Using $m\phi_0 \sim T_\mathrm{rec}^2$ and evaluating at recombination, the corresponding physical wavelength is

$$\lambda_c \sim \tilde{g}_{\phi\gamma}^{-1} \times 10^{-6}\,\mathrm{Mpc}, \tag{2.8}$$

which is enormously larger than the scales relevant to DCBH formation.

The instability sets in right after recombination when the electric conductance of the plasma drops abruptly (the plasma damping term $\sigma A_k'$ becomes negligible). The resulting field is *helical* because the two polarisations obey different equations—a key distinction from the scalar coupling studied in Kamali & Brandenberger.

2.5 Two Regimes: Tachyonic vs Parametric Resonance

- **Tachyonic resonance** (broad-band, efficient): active when $\tilde{g}_{\phi\gamma}\, m_{20}^{-1} > 1$. All modes $k < k_\mathrm{c}$ grow simultaneously.
- **Parametric resonance** (narrow-band): dominant when $\tilde{g}_{\phi\gamma}\, m_{20}^{-1} < 1$. The instability band is centred at $k \sim am$, corresponding to $\lambda \sim m_{20}^{-1} \times 10^{-6}\,\mathrm{Mpc}$.

In either case, the wavelength of the generated magnetic field is many orders of magnitude larger than the scales of interest for the DCBH calculation ($k \sim 10\,\mathrm{eV}$), so the magnetic field is effectively *uniform* on those scales. This is a crucial simplification.

2.6 Amplitude of the Background Magnetic Field

The resonance shuts off when a fraction $F \ll 1$ of the dark matter energy has transferred to the gauge field. Energy conservation gives

$$B \sim F^{1/2}\,\mathrm{G} \simeq 1\,\mathrm{G} \quad (\text{taking } F \sim 1). \tag{2.9}$$

For the present paper, $B \sim 1\,\mathrm{G}$ is taken as the input amplitude of the cosmological magnetic field on scale k_c, generated by the ALP mechanism. The calculation proceeds with this as a given background.

3 Spacetime Setup and Field Definitions

3.1 FRW Metric in Conformal Time

We work with the flat Friedmann-Lemaître-Robertson-Walker (FRW) metric written in conformal time η with signature $(+,-,-,-)$:

$$ds^2 = a^2(\eta)\left(d\eta^2 - \delta_{ij}\,dx^i dx^j\right), \tag{3.1}$$

so that $g_{00} = a^2$, $g_{ij} = -a^2\delta_{ij}$, and $\sqrt{-g} = a^4$.

3.2 Comoving Observer and Electromagnetic Fields

The four-velocity of a comoving observer is

$$u^\mu = \left(a^{-1},\, 0,\, 0,\, 0\right), \qquad u_\mu = (a,\, 0,\, 0,\, 0), \qquad u^\mu u_\mu = 1. \tag{3.2}$$

In Coulomb gauge ($A_0 = 0$, $\partial_i A^i = 0$), the field strength tensor has components

$$F_{0i} = \partial_0 A_i = A_i', \qquad F_{ij} = \partial_i A_j - \partial_j A_i. \tag{3.3}$$

The electric and magnetic fields measured by the comoving observer are defined by $E_\mu = F_{\mu\nu}u^\nu$ and $B^\mu = \frac{1}{2}\varepsilon^{\mu\nu\alpha\beta}u_\nu F_{\alpha\beta}$.

Derivation 3.1 (Electric field of the comoving observer)**.** For the spatial components:

$$E_i = F_{i\nu}u^\nu = F_{i0}\,u^0 = -F_{0i}\cdot a^{-1} = -\frac{A_i'}{a}. \tag{3.4}$$

So the physical 3-vector electric field has components $\mathcal{E}_i = -A_i'/a$. We write $E_\mu = (0, -A_i'/a) = a(0, \boldsymbol{\mathcal{E}})$ where $\mathcal{E}_i = -A_i'/a^2$.

Derivation 3.2 (Magnetic field of the comoving observer)**.** Using $\varepsilon^{0ijk} = \varepsilon^{ijk}/a^4$ (Levi-Civita *tensor* in curved space, where $\varepsilon^{\mu\nu\alpha\beta} = \bar{\varepsilon}^{\mu\nu\alpha\beta}/\sqrt{-g}$ with $\bar{\varepsilon}^{0123} = +1$):

$$B^i = \frac{1}{2}\varepsilon^{i\nu\alpha\beta}u_\nu F_{\alpha\beta} = \frac{1}{2}\varepsilon^{i0jk}u_0 F_{jk} = \frac{a}{2}\cdot\frac{\varepsilon^{ijk}}{a^4}\cdot F_{jk} = \frac{1}{2a^3}\varepsilon^{ijk}F_{jk} = \frac{1}{a^3}\varepsilon^{ijk}\partial_j A_k. \tag{3.5}$$

Thus $\mathcal{B}^i = B^i/\cdots$ and, with the lowering convention, $\mathcal{B}_i = -\varepsilon_{ijk}\partial_j A_k/a^3$.

3.3 Rescaled Fields

It is very convenient to introduce the rescaled fields

$$\boxed{\mathcal{E}_i \equiv a^2 E_i^{\text{phys}}, \qquad \mathcal{B}_i \equiv a^2 B_i^{\text{phys}}}, \tag{3.6}$$

where E_i^{phys} and B_i^{phys} are the physical (observer-frame) field components. In highly conducting plasma these physical fields decay as a^{-2}, so the rescaled fields are *approximately conserved*.

The rescaled fields satisfy the simple relations (shown in Appendix A)

$$\mathcal{E}_i = -A_i', \qquad \mathcal{B}_i = -\varepsilon_{ijk}\,\partial_j A_k. \tag{3.7}$$

Note
The relation $\mathcal{E}_i = -A_i'$ means that the rescaled electric field is simply the conformal-time derivative of the gauge potential, *with no factor of* a. This is one of the main advantages of working with rescaled fields: the equations of motion take the same form as in Minkowski space.

The energy density of the magnetic field is

$$\rho_B = \frac{1}{2a^4}\mathcal{B}\cdot\mathcal{B} = \frac{1}{2}\left|\boldsymbol{B}_{\text{phys}}\right|^2. \tag{3.8}$$

4 Equation of Motion for Secondary Gauge Field Perturbations

4.1 Full Equations of Motion

Varying the action with $\mathcal{L}$ as in equation (2.1) with respect to ϕ and A^μ yields the Klein-Gordon and modified Maxwell equations:

$$\frac{1}{\sqrt{-g}}\partial_\mu\big(\sqrt{-g}\,\partial^\mu\phi\big) + m^2\phi = \frac{1}{4}g_{\phi\gamma}\,F_{\mu\nu}\tilde{F}^{\mu\nu}, \tag{4.1}$$

$$\partial_\mu\big(\sqrt{-g}\,F^{\mu\nu}\big) = g_{\phi\gamma}\,\partial_\mu\Big(\sqrt{-g}\,\phi\,\tilde{F}^{\mu\nu}\Big)\,. \tag{4.2}$$

4.2 Decomposition into Background and Fluctuation

We split all fields into a spatially homogeneous background (denoted with a bar) and a small perturbation (denoted with δ):

$$\phi(\eta,\boldsymbol{x}) = \bar{\phi}(\eta) + \delta\phi(\eta,\boldsymbol{x}), \qquad F_{\mu\nu} = \bar{F}_{\mu\nu} + \delta F_{\mu\nu}. \tag{4.3}$$

The background part $\bar{\phi} = \phi_0\cos(m\eta)$ satisfies its own equation (4.1), generating the cosmological magnetic field $\bar{F}_{\mu\nu}$ via the mechanism described in Chapter 2. We treat this as a known, uniform background.

The secondary photons are described by the perturbation $\delta F_{\mu\nu}$, characterised by the gauge field perturbation A_i (a new field, distinct from the background). We work to *linear order* in $\delta\phi$ and A_i.

4.3 Linearised Equation of Motion

Inserting the decomposition into (4.2) and retaining only first-order terms, the $\nu = i$ (spatial) component of the modified Maxwell equation becomes (in Coulomb gauge):

$$\boxed{A_i'' - \nabla^2 A_i = S_i(\eta,\boldsymbol{x})}, \tag{4.4}$$

with the source

$$\boxed{S_i = -g_{\phi\gamma}\,(\delta\phi)'\,\mathcal{B}_i + g_{\phi\gamma}\,\varepsilon_{ijk}\,\partial_j(\delta\phi)\,\mathcal{E}_k}. \tag{4.5}$$

Here $\mathcal{B}_i$ and $\mathcal{E}_k$ are the rescaled background fields from equation (3.7). A complete derivation of equations (4.4)–(4.5) from first principles is given in Appendix A.

4.4 Physical Interpretation of the Source Terms

- **First term** $-g_{\phi\gamma}(\delta\phi)'\mathcal{B}_i$: The time-varying ALP fluctuation $\delta\phi$ acts like a time-varying "dielectric constant" in the magnetic background, driving a radiation field proportional to $(\dot{\delta\phi})$ and the background magnetic field. This is the dominant term when the electric background $\mathcal{E} = 0$.

- **Second term** $g_{\phi\gamma}\,\varepsilon_{ijk}\partial_j(\delta\phi)\mathcal{E}_k$: Spatial gradients of $\delta\phi$ couple to the background electric field. For a purely magnetic background ($\mathcal{E} = 0$), this term vanishes. The paper sets $\mathcal{E} = 0$ for the background, retaining only the first term.

Note
Note the crucial scale separation: the background $\mathcal{B}$ is on scales $\lambda \sim k_{\rm c}^{-1} \sim \tilde{g}_{\phi\gamma}^{-1} \times 10^{-6}\,\mathrm{Mpc}$, while the DM fluctuation $\delta\phi(k)$ is on sub-Hubble scales from the primordial perturbation spectrum, covering a range down to $k \sim 10\,\mathrm{eV}$ (the LW band). The secondary photons inherit the wavenumber of $\delta\phi$ (since B is taken uniform), which is why this mechanism can directly produce LW photons.

4.5 Simplification: Uniform Background, No Background Electric Field

Since the magnetic field is produced on scales $\lambda_c \gg$ (scales of interest), we treat $\mathcal{B}$ as spatially uniform, $\mathcal{B}(\eta, \boldsymbol{x}) \approx \mathcal{B}(\eta)$. For a uniform, time-evolving magnetic field in Coulomb gauge, the background electric field vanishes: $\mathcal{E} = 0$. After recombination the field redshifts as $\mathcal{B} \sim a^{-2}$. The source simplifies to

$$S_i(k, \eta) = -g_{\phi\gamma}\,(\delta\phi)'(k, \eta)\,\mathcal{B}_i(\eta), \tag{4.6}$$

where the Fourier transform has been taken: $A_i''(k) + k^2 A_i(k) = -g_{\phi\gamma}\,(\delta\phi)'(k)\,\mathcal{B}_i(\eta)$.

5 Green's Function Solution

5.1 Initial Value Problem

After recombination, when the plasma conductance drops, we treat the secondary gauge field as starting from rest:

$$A_i(k, \eta_{\rm rec}) = 0, \qquad A_i'(k, \eta_{\rm rec}) = 0. \tag{5.1}$$

The problem is then to solve the forced oscillator equation

$$A_i''(k, \eta) + k^2 A_i(k, \eta) = S_i(k, \eta), \tag{5.2}$$

with zero initial data (5.1).

5.2 Construction of the Green's Function

The Green's function $G(k; \eta, \tau)$ for equation (5.2) satisfies

$$G''(k; \eta, \tau) + k^2 G(k; \eta, \tau) = \delta(\eta - \tau), \tag{5.3}$$

and vanishes for $\eta < \tau$ (causality). The homogeneous solutions are $f_1 = \cos(k\eta)$ and $f_2 = \sin(k\eta)$, with Wronskian

$$W = f_1 f_2' - f_2 f_1' = k\cos^2(k\eta) + k\sin^2(k\eta) = k. \tag{5.4}$$

Derivation 5.1 (Green's function by variation of parameters)**.** The Green's function is

$$G(k; \eta, \tau) = \frac{f_1(k; \eta) f_2(k; \tau) - f_2(k; \eta) f_1(k; \tau)}{W(k; \tau)}. \tag{5.5}$$

Inserting $f_1 = \cos(k\eta)$, $f_2 = \sin(k\eta)$, and $W = k$:

$$\begin{aligned} G(k; \eta, \tau) &= \frac{\cos(k\eta)\sin(k\tau) - \sin(k\eta)\cos(k\tau)}{k} \\ &= \frac{-\sin(k\eta - k\tau)}{k} = \frac{\sin(k(\tau - k\eta))}{k} \\ &= \frac{\sin\big(k(\eta - \tau)\big)}{k} \cdot (-1) \end{aligned} \tag{5.6}$$

Wait—let us be careful about signs. The combination $f_1(k; \eta) f_2(k; \tau) - f_2(k; \eta) f_1(k; \tau) = \cos(k\eta)\sin(k\tau) - \sin(k\eta)\cos(k\tau) = -\sin(k(\eta - \tau))$. Dividing by $W = k$ and noting that we need $G(\eta, \tau) = 0$ for $\eta < \tau$, the correct retarded Green's function is

$$G(k; \eta, \tau) = \frac{1}{k}\sin\big(k(\eta - \tau)\big) \quad (\eta \geq \tau), \qquad G = 0 \quad (\eta < \tau). \tag{5.7}$$

One verifies: $G''(\eta, \tau) + k^2 G(\eta, \tau) = \delta(\eta - \tau)$ since the function is continuous at $\eta = \tau$ (where $G = 0$) and has a unit jump in its first derivative (from 0 to $k^{-1} \cdot k = 1$). $\square$

5.3 Formal Solution

By the method of Green's functions, the solution satisfying the initial conditions (5.1) is

$$\boxed{A_i(k,\eta) = \int_{\eta_{\rm rec}}^{\eta} G(k;\eta,\tau)\, S_i(k,\tau)\, d\tau = \int_{\eta_{\rm rec}}^{\eta} \frac{\sin[k(\eta-\tau)]}{k}\, S_i(k,\tau)\, d\tau.} \tag{5.8}$$

Inserting the simplified source (4.6):

$$A_i(k,\eta) = -\frac{g_{\phi\gamma}\mathcal{B}_i}{k} \int_{\eta_{\rm rec}}^{\eta} \sin\big[k(\eta-\tau)\big] \cdot (\delta\phi)'(k,\tau)\, d\tau. \tag{5.9}$$

5.4 Redshifting of the Background Field

After recombination the background magnetic field $\mathcal{B}$ redshifts as a^{-2}:

$$\mathcal{B}(\tau) = \mathcal{B}(\eta_{\rm rec}) \left[\frac{a(\eta_{\rm rec})}{a(\tau)}\right]^2 . \tag{5.10}$$

This τ-dependence sits inside the integral and provides a suppression at late times—an important point for estimating where the photon production is concentrated.

6 Cosmological Perturbations and the Source

6.1 Scale-Invariant Primordial Power Spectrum

We assume a Harrison-Zel'dovich (scale-invariant) spectrum of primordial perturbations. The dimensionless power spectrum of density perturbations, evaluated when scale k crosses the Hubble radius at time $t_i(k)$, is

$$\left(\frac{\delta\rho}{\rho}\right)^2 k^3 \bigg|_{k=aH} = \mathcal{A}, \tag{6.1}$$

where $\mathcal{A} \simeq 10^{-9}$ is set by the amplitude of CMB fluctuations.

Scales of interest for DCBH formation (down to $k \sim 10\,\mathrm{eV}$ at recombination) entered the Hubble radius long before recombination and were subsequently frozen until matter-radiation equality (up to logarithmic corrections). Hence the power spectrum is flat (independent of k) over the scales of interest at η_{rec}.

6.2 Dark Matter Field Fluctuation

The ALP field ϕ carries the dark matter energy, so density fluctuations $\delta\rho/\rho$ in the dark matter are inherited by ϕ. For a quadratic potential $V = \frac{1}{2}m^2\phi^2$, the energy density is $\rho_\phi = \frac{1}{2}m^2\phi^2$, and the fractional fluctuation is

$$\frac{\delta\rho_\phi}{\rho_\phi} = \frac{\delta(\phi^2)}{\phi^2} \approx \frac{2\phi_0\,\delta\phi}{\phi_0^2} = \frac{2\,\delta\phi}{\phi_0}. \tag{6.2}$$

Thus

$$\delta\phi(k) = \frac{\phi_0}{2} \cdot \frac{\delta\rho}{\rho}(k). \tag{6.3}$$

From the scale-invariant spectrum (6.1):

$$\frac{\delta\rho}{\rho}(k) = \mathcal{A}^{1/2}k^{-3/2}, \tag{6.4}$$

so

$$\delta\phi(k) = \frac{1}{2}\phi_0\,\mathcal{A}^{1/2}\,k^{-3/2}. \tag{6.5}$$

Note
The factor $k^{-3/2}$ means that the field fluctuation amplitude *increases* toward large scales (small k). However, the photon energy density will involve an extra factor of k^2 from the dispersion relation and factors of k from phase space, so the final observable flux turns out to decrease toward large scales.

6.3 Time Dependence of the Source

The source (equation (4.6)) involves the time derivative of $\delta\phi$. After recombination, sub-Hubble fluctuations in the dark matter field oscillate at frequency m (the ALP mass), so:

$$\delta\phi(k,\eta) \approx \delta\phi(k)\,\cos(m\eta), \qquad (\delta\phi)'(k,\eta) \approx -m\,\delta\phi(k)\,\sin(m\eta). \tag{6.6}$$

However, in the integral (5.9), the dominant contribution comes from the oscillating factors $\sin[k(\eta-\tau)]$ and $(\delta\phi)'(k,\tau)$ together with the redshifting factor $(a_{\rm rec}/a(\tau))^2$. The last factor strongly suppresses contributions from times $\tau \gg \eta_{\rm rec}$, so the photon production is concentrated near recombination.

7 Computing the Secondary Photon Spectrum

7.1 Evaluating the Integral: Order-of-Magnitude Estimate

We now estimate the amplitude $A_i(k,\eta)$ from equation (5.9). Inserting the source (4.6) and the time-dependence (6.6) (taking $(\delta\phi)' = -m\,\delta\phi(k)\cos(m\tau)$ schematically):

$$A_i(k,\eta) \sim \frac{g_{\phi\gamma}\mathcal{B}_i(\eta_{\rm rec})}{k}\,\delta\phi(k)\int_{\eta_{\rm rec}}^{\eta} \sin\!\big[k(\eta-\tau)\big]\cos\!\big[k(\tau-\eta_{\rm rec})\big]\left[\frac{a(\eta_{\rm rec})}{a(\tau)}\right]^2 d\tau. \tag{7.1}$$

The integral is evaluated by noting:

- The factor $(a_{\rm rec}/a(\tau))^2$ suppresses contributions from $\tau \gg \eta_{\rm rec}$; the integration is effectively cut off near $\tau \sim \eta_{\rm rec}$.
- Changing variables to $x \equiv k(\tau-\eta_{\rm rec})$, the integral becomes $k^{-1}\int_0^{k(\eta-\eta_{\rm rec})} \sin[k(\eta-\eta_{\rm rec})-x]\cos(x)\;\times\;(\text{suppression})\,dx$.
- By dimensional analysis, and since the suppression factor effectively limits the integration region, the integral is of order k^{-1}.

Therefore, to order of magnitude:

$$A_i(k,\eta) \sim \frac{g_{\phi\gamma}\,\mathcal{B}\,\delta\phi(k)}{k^2}, \tag{7.2}$$

where $\mathcal{B} \equiv |\mathcal{B}_i(\eta_{\rm rec})|$.

7.2 Energy Density Spectrum

The comoving energy density in the secondary photons per unit wavenumber interval is

$$\rho_A(k) \sim k^2|A_k(\eta)|^2, \tag{7.3}$$

where the factor k^2 comes from the $(\partial A)^2$ kinetic term in the electromagnetic Lagrangian (schematically $\rho \sim k^2|A|^2$ from $|\nabla A|^2 \sim k^2|A|^2$).

Substituting (7.2) and (6.5):

$$\rho_A(k) \sim k^2\cdot\left(\frac{g_{\phi\gamma}\,\mathcal{B}\,\delta\phi(k)}{k^2}\right)^2 = g_{\phi\gamma}^2\,\mathcal{B}^2\cdot\frac{|\delta\phi(k)|^2}{k^2} \sim g_{\phi\gamma}^2\,\mathcal{B}^2\cdot\frac{\phi_0^2\,\mathcal{A}}{4}\cdot\frac{k^{-3}}{k^2} = \frac{g_{\phi\gamma}^2\,\mathcal{B}^2\,\phi_0^2\,\mathcal{A}}{4}\cdot k^{-3}. \tag{7.4}$$

7.3 Photon Number Flux

The photon number flux at wavenumber k (number of photons per unit volume per unit time per unit frequency interval) is related to the energy density by

$$\Phi_A(k) \sim \frac{k^3\,\rho_A(k)}{k} = k^2\,\rho_A(k), \tag{7.5}$$

where the factor k^3 comes from the phase space volume $d^3k/(2\pi)^3 \sim k^3$ and the extra k^{-1} converts energy density to number density (dividing by photon energy k). Using (7.4):

$$\Phi_A(k) \sim g_{\phi\gamma}^2\,\mathcal{B}^2\,\mathcal{A}\,\phi_0^2 \cdot k^{-1}. \tag{7.6}$$

The flux scales as k^{-1}: lower-energy photons are produced more abundantly. This is physically reasonable since the source $\delta\phi(k) \propto k^{-3/2}$ favours large-scale (small-k) modes.

7.4 Inserting Numerical Values

We now evaluate the flux at $k \sim 10\,\mathrm{eV}$ (the Lyman-Werner band).

Lyman-Werner photon flux (parametric form)
Using $g_{\phi\gamma} = \tilde{g}_{\phi\gamma} \times 10^{-10}\,\mathrm{GeV}^{-1}$, $\mathcal{B} \sim 1\,\mathrm{G} = 1.95 \times 10^{-2}\,\mathrm{eV}^2$, $\mathcal{A} \sim 10^{-9}$, $\phi_0 \sim T_{\rm rec}^2/m$, and $k \sim 10\,\mathrm{eV}$:

$\sim \frac{T_{\rm rec}^4}{m^2} = \frac{(0.3\,\mathrm{eV})^4}{(m_{20}\times 10^{-20}\,\mathrm{eV})^2} \sim \frac{8.1\times 10^{-3}\,\mathrm{eV}^4}{m_{20}^2\times 10^{-40}\,\mathrm{eV}^2} = \frac{8.1\times 10^{37}}{m_{20}^2}\,\mathrm{eV}^2,$
g$_{\phi\gamma}^2$
$= {}^{2}_{\phi\gamma} \times 10^{-20}\,\mathrm{GeV}^{-2} = \tilde{g}_{\phi\gamma}^2 \times 10^{-20} \times 10^{-18}\,\mathrm{eV}^{-2} = \tilde{g}_{\phi\gamma}^2 \times 10^{-38}\,\mathrm{eV}^{-2}.$

Hence

$$\begin{aligned}
\Phi_A(k{=}10\,\mathrm{eV}) &\sim \tilde{g}_{\phi\gamma}^2 \times 10^{-38}\,\mathrm{eV}^{-2} \times (3.8\times 10^{-4}\,\mathrm{eV}^4) \times 10^{-9} \times \frac{8.1\times 10^{37}}{m_{20}^2}\,\mathrm{eV}^2 \times (10\,\mathrm{eV})^{-1} \\
&\sim \frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} \times 10^{-38} \times 3.8\times 10^{-4} \times 10^{-9} \times 8.1\times 10^{37} \times 10^{-1}\,\mathrm{eV}^3 \\
&\sim \frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} \times 1.2\times 10^{-14}\,\mathrm{eV}^3 \sim \frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} \times 10^{-39}\,\mathrm{GeV}^3,
\end{aligned} \tag{7.7}$$

where we used $1\,\mathrm{GeV}^3 = 10^{27}\,\mathrm{eV}^3$.

This matches the result quoted as equation (23) of Kushwaha & Brandenberger.

8 The DCBH Formation Criterion

8.1 The Lyman-Werner Flux Requirement

For the Direct Collapse Black Hole scenario to work, the flux of LW photons at $k_{\rm LW} \sim 10\,\mathrm{eV}$ must be sufficient to dissociate H_2 and prevent cloud fragmentation. Translating the critical LW flux $J_{\rm LW,crit}$ into the units of Φ_A used here, the requirement is

$$\Phi_A(k_{\rm LW}) \gtrsim \Phi_{\rm LW,crit} \sim 10^{-44}\,\mathrm{GeV}^3. \tag{8.1}$$

(The conversion between J_{21} units and GeV^3 involves the photon energy $k_{\rm LW} \sim 10\,\mathrm{eV}$ and the solid-angle and frequency integrals.)

8.2 The Sufficient Condition

Comparing the computed flux (7.7) with the criterion (8.1):

$$\frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} \times 10^{-39}\,\mathrm{GeV}^3 \;\gtrsim\; 10^{-44}\,\mathrm{GeV}^3, \tag{8.2}$$

which gives

$$\boxed{\frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} \gtrsim 10^{-5}.} \tag{8.3}$$

DCBH Criterion

The LW photon flux from ALP dark matter oscillations interacting with the cosmological magnetic field is sufficient to enable DCBH formation provided

$$\tilde{g}_{\phi\gamma}^2\, m_{20}^{-2} \gtrsim 10^{-5}. \tag{8.4}$$

This can be satisfied for values of $g_{\phi\gamma}$ and m that are consistent with all other observational constraints (see Chapter 9).

8.3 Why This Mechanism is Attractive

i. **No separate LW source needed.** Traditional DCBH models require a bright galaxy in close proximity to the collapsing halo. The ALP mechanism provides a spatially uniform cosmological LW background.

ii. **Direct channel, not a cascade.** Unlike the companion proposal in [3] where an infrared-to-UV photon cascade is invoked, the mechanism here directly sources photons at $k \sim 10\,\mathrm{eV}$ via the primordial fluctuations on that scale.

iii. **Consistency.** The same ALP that generates the cosmological magnetic field also generates the LW photons through a simple coupling to the density fluctuations.

9 Constraints from ARCADE2 and EDGES

9.1 The Excess Radio Background

Two experiments have reported (contested) detections of an excess of radio photons above the expected CMB blackbody:

- **ARCADE2**: measures at frequencies 3–90 GHz, reporting an excess temperature $\delta T/T_{\rm CMB} \lesssim 4 \times 10^{-4}$ at $\omega/T_{\rm CMB} \sim 0.2$.
- **EDGES**: measures a 21 cm absorption feature at 78 MHz ($\omega/T_{\rm CMB} \sim 1.4 \times 10^{-3}$), with an excess $\delta T/T_{\rm CMB} < 1$.

Both detections are disputed and the physical origin of the excess remains open. Kushwaha & Brandenberger use these as *upper bounds* on any additional photon flux, providing constraints on the ALP coupling.

9.2 Brightness Temperature Formula

The brightness temperature $\delta T(\omega)$ of a photon flux $\Phi_A(\omega)$ is related by the Rayleigh-Jeans formula:

$$\delta T(\omega) = \frac{4\pi^2}{\omega^2}\,\Phi_A(\omega). \tag{9.1}$$

This follows from the thermal radiation formula in the Rayleigh-Jeans limit ($\omega \ll T$): the specific intensity $I_\nu = 2k_B T\nu^2/c^2$, so $\delta T = c^2\Phi_A/(2k_B\nu^2) \propto \Phi_A/\omega^2$.

Derivation 9.1 (ARCADE2 constraint)**.** The ARCADE2 bound is

$$\frac{\delta T(\omega_{\rm ARCADE2})}{T_{\rm CMB}} < 4 \times 10^{-4}, \quad \frac{\omega_{\rm ARCADE2}}{T_{\rm CMB}} \sim 0.2. \tag{9.2}$$

Substituting the brightness temperature formula (9.1):

$$\frac{4\pi^2}{\omega_{\rm A}^2}\,\Phi_A(\omega_{\rm A}) < 4 \times 10^{-4}\,T_{\rm CMB}. \tag{9.3}$$

Now we use the flux formula (7.6): $\Phi_A(\omega) \propto \omega^{-1}$. The formula (7.7) was derived at $k = 10\,$eV (LW band). At the ARCADE2 frequency $\omega_{\rm A} \sim 0.2\,T_{\rm CMB} \sim 4.6 \times 10^{-5}\,$eV, the same formula applies (the production mechanism is the same for any photon wavenumber k smaller than $k_{\rm c}$):

$$\Phi_A(\omega) \sim \frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} \times 10^{-39}\,{\rm GeV}^3 \times \frac{k_{\rm LW}}{\omega}. \tag{9.4}$$

Inserting into (9.3) and using $T_{\rm CMB} \simeq 2.3 \times 10^{-4}\,\text{eV}$:

$$\frac{4\pi^2}{\omega_{\rm A}^2} \cdot \frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} \times 10^{-39}\,\text{GeV}^3 \times \frac{k_{\rm LW}}{\omega_{\rm A}} < 4 \times 10^{-4}\,T_{\rm CMB}. \tag{9.5}$$

Working through the numerical factors (with $\omega_{\rm A} \sim 0.2\,T_{\rm CMB}$ and all quantities in GeV), the ARCADE2 condition becomes

$$\boxed{\frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} < 4 \times 10^5.} \tag{9.6}$$

9.3 EDGES Constraint

The EDGES measurement at $\omega_{\rm E}/T_{\rm CMB} \sim 1.4 \times 10^{-3}$ gives $\delta T/T_{\rm CMB} < 1$, which translates (via an analogous calculation) to the slightly tighter bound:

$$\frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} < 5 \times 10^4. \tag{9.7}$$

9.4 Summary of Constraints

Two-sided bounds on the parameter combination

Combining the DCBH lower bound (8.3) with the ARCADE2/EDGES upper bounds (9.6)–(9.7):

$$10^{-5} \lesssim \frac{\tilde{g}_{\phi\gamma}^2}{m_{20}^2} \lesssim 5 \times 10^4. \tag{9.8}$$

This represents a viable parameter window spanning about nine orders of magnitude. The bounds are satisfied for a wide range of ALP masses consistent with ultralight dark matter constraints ($m_{20} \gtrsim 10$).

10 Parameter Space and Discussion

10.1 Consistency with Existing ALP Constraints

The coupling constant is bounded independently by astrophysical observations:

$$g_{\phi\gamma} < 10^{-9}\, m_{20}\, \mathrm{GeV}^{-1} \quad \Rightarrow \quad \tilde{g}_{\phi\gamma} < 10\, m_{20}, \quad \Rightarrow \quad \frac{\tilde{g}^2_{\phi\gamma}}{m^2_{20}} < 100. \tag{10.1}$$

Comparing with the ARCADE2 bound $\tilde{g}^2_{\phi\gamma}/m^2_{20} < 4 \times 10^5$: the astrophysical bound is actually *more* restrictive, but both are consistent with the window found above. The DCBH condition $\tilde{g}^2_{\phi\gamma}/m^2_{20} \gtrsim 10^{-5}$ requires only $\tilde{g}_{\phi\gamma} \gtrsim 3 \times 10^{-3}\, m_{20}$, which is easily satisfied.

10.2 The Role of the Fraction F

Throughout the calculation we took $B \sim 1\,\mathrm{G}$, corresponding to $F \sim 1$ (all dark matter energy converts to the magnetic field). If only a fraction $F \ll 1$ converts, then $B \sim F^{1/2}\,\mathrm{G}$ and the photon flux scales as $B^2 \propto F$. The LW condition (8.3) and ARCADE2 bound (9.6) are both proportional to F, so they remain a consistent two-sided window in the $(F, \tilde{g}^2_{\phi\gamma}/m^2_{20})$ plane.

10.3 Generality: Different Origins of B

An important strength of the mechanism is that it does not require the magnetic field to be generated by the ALP mechanism. Any source of a cosmological magnetic field $B \sim 1\,\mathrm{G}$ on large scales (e.g. from inflationary magnetogenesis, or the scalar DM mechanism of Kamali & Brandenberger) would produce a similar secondary photon flux when coupled to the DM fluctuations via $g_{\phi\gamma}\phi F \wedge F$.

With minor modifications, the mechanism also works if the dark matter is a scalar field coupled to F^2 (as studied in the companion paper), since the relevant parameter is $B^2\phi_0^2 g^2_{\phi\gamma}/k$, and the same order-of-magnitude estimates apply.

10.4 Small-Scale Enhancement of the Primordial Spectrum

The calculation assumed a flat (scale-invariant) spectrum of primordial perturbations down to scales $k \sim 10\,\mathrm{eV}$. If the spectrum is enhanced on small scales (as postulated in primordial black hole formation scenarios), the LW flux would be proportionally increased. However, such an enhancement on ARCADE2 scales would be in conflict with the observed bounds—providing a new and interesting constraint on small-scale power.

11 Comparison of Scalar vs. Pseudoscalar Coupling

11.1 Side-by-Side Comparison

Property	Pseudoscalar (ALP), this paper	Scalar (dilaton), Kamali & Brandenberger
Coupling	$g_{\phi\gamma}\phi F_{\mu\nu}\tilde{F}^{\mu\nu}$	$I(\phi)F_{\mu\nu}F^{\mu\nu}$
Helicity of B	Helical	Non-helical
Mode eq.	$\ddot{A}_k^{\pm} + (k^2 \pm g_{\phi\gamma}\dot{\phi}k)A_k^{\pm} = 0$	$v_k'' + (k^2 - (\sqrt{I})''/\sqrt{I})v_k = 0$
k_c	$\sim a g_{\phi\gamma} m \phi_0$	$\sim m\sqrt{\gamma}$
Spectrum slope	$n = 3/2$	$n = 1$
Fine-structure const.	Not directly affected	Oscillates at frequency m
Signature	Helical CMB polarisation	Oscillating α
LW photon channel	Yes (this paper)	Possible (minor modification)
$B_0(1\,\mathrm{Mpc})$	$\sim 10^{-15}\,\mathrm{G}$	$\sim 10^{-8}\,\mathrm{G}$

Table 11.1: Comparison of scalar and pseudoscalar dark matter magnetogenesis mechanisms.

11.2 Which is Better for Magnetogenesis?

In terms of the amplitude of magnetic field on 1 Mpc scales today, the scalar (dilaton) coupling produces a larger field ($\sim 10^{-8}\,\mathrm{G}$ vs. $\sim 10^{-15}\,\mathrm{G}$ for the pseudoscalar), mainly because the non-helical inverse cascade (slope $n = 1$) transfers power to large scales more efficiently than the helical cascade ($n = 3/2$). Both, however, exceed the observational lower bound of $10^{-17}\,\mathrm{G}$.

11.3 Which is Better for DCBH Formation?

The pseudoscalar coupling has an advantage here: the $\phi F \wedge F$ term directly couples the ALP oscillation to electromagnetism in a way that sources both the background field *and* the secondary photons through a single parameter $g_{\phi\gamma}$. The resulting LW flux is then bounded from both above (ARCADE2/EDGES) and below (DCBH criterion), providing a predictive and testable parameter window.

12 Open Questions and Limitations

12.1 Magnetic Field Inhomogeneities

The calculation treated the background magnetic field B as spatially uniform. In reality, the tachyonic resonance generates a field that, while peaked at $k_{\rm c}$, has some spatial variation. Including this would replace the source term by a convolution

$$S_i(k,\eta) \sim g_{\phi\gamma} \int d^3k' \, (\delta\phi)'(k') \, \mathcal{B}(k-k'). \tag{12.1}$$

This convolution couples different Fourier modes and could lead to additional photon production on scales between $k_{\rm c}$ and $k_{\rm LW}$.

12.2 Spectral Tilt

A small red tilt $n_s \approx 0.965$ in the primordial power spectrum has been neglected. Including it modifies $\delta\phi(k) \propto k^{-(3+n_s-1)/2}$ at the sub-percent level—negligible for the current order-of-magnitude analysis.

12.3 Scale Dependence of Perturbation Spectrum below Equality

The scale-invariant spectrum holds on scales larger than the Jeans length at matter-radiation equality. On much smaller scales (including $k \sim 10\,\text{eV}$ at recombination), transfer function corrections suppress the matter power spectrum. Whether the spectrum genuinely extends to $k \sim 10\,\text{eV}$ at $\eta_{\rm rec}$ depends on the model for early-universe perturbations; it holds for inflation with a smooth inflaton potential but may fail in alternatives.

12.4 Back-Reaction on the ALP Field

The production of secondary photons drains energy from the ALP oscillation, reducing ϕ_0 and hence $\delta\phi$. For the flux computed here ($\Phi_A \ll \rho_\phi$), back-reaction is negligible. A more careful treatment would involve evolving the coupled system self-consistently.

12.5 Plasma Effects Post-Recombination

A residual free-electron fraction $x_e \sim 10^{-3}$ persists after recombination. The plasma conductivity $\sigma \sim x_e T_e$ adds a damping term $\sigma A_k'$ to the mode equation, potentially suppressing the generation of secondary photons on scales smaller than the plasma mean free path. This has not been computed.

13 Conclusions

The central result of Kushwaha & Brandenberger (2026) is that the cosmological magnetic field generated by ALP dark matter oscillations, when coupled to the primordial dark matter density fluctuations via the $g_{\phi\gamma}\phi F \wedge F$ term, produces a flux of Lyman-Werner photons sufficient to enable Direct Collapse Black Hole formation.

The chain of logic runs as follows:

1. An oscillating ALP with mass $m \sim 10^{-20}$–$10^{-10}\,\mathrm{eV}$ and coupling $g_{\phi\gamma}$ to electromagnetism generates cosmological magnetic fields of amplitude $B \sim 1\,\mathrm{G}$ at $k = k_{\mathrm{c}}$ right after recombination via tachyonic resonance.

2. The same coupling drives a secondary photon production process: $\delta\phi(k) + B \to \delta A_k$. The dark matter density fluctuation $\delta\phi(k)$ on scale k serves as the source; the background B acts as a catalyst.

3. The induced gauge field is computed via the Green's function of the wave equation, with initial conditions set at recombination. The dominant contribution comes from near the recombination epoch.

4. The photon number flux at wavenumber k is $\Phi_A(k) \sim g_{\phi\gamma}^2 B^2 \mathcal{A}\phi_0^2/k$, scaling as k^{-1}.

5. Evaluating at the LW band ($k \sim 10\,\mathrm{eV}$) and inserting the ALP parameters: $\Phi_A \sim \tilde{g}_{\phi\gamma}^2 m_{20}^{-2} \times 10^{-39}\,\mathrm{GeV}^3$.

6. The DCBH criterion (Φ_A larger than a critical LW flux) requires $\tilde{g}_{\phi\gamma}^2/m_{20}^2 \gtrsim 10^{-5}$.

7. Consistency with ARCADE2 (upper bound on excess radio photons) requires $\tilde{g}_{\phi\gamma}^2/m_{20}^2 \lesssim 4 \times 10^5$. A viable parameter window therefore exists spanning nine decades.

The mechanism is notable for several reasons. It provides a *cosmological*, model-independent source of Lyman-Werner radiation that does not require proximity to a specific galaxy. It links the origin of supermassive black holes to the nature of dark matter, making it testable: future precision measurements of the diffuse radio background (from experiments like SKA) or 21 cm cosmology could constrain the parameter space. The mechanism also places new constraints on small-scale power in the primordial perturbation spectrum: an enhanced spectrum on ARCADE2 scales would overproduce radio photons.

A Full Derivation of the Secondary Photon EOM

This appendix provides the complete derivation of the wave equation for the secondary gauge field, as presented in the appendix of Kushwaha & Brandenberger.

A.1 Metric and Christoffel Symbols

With the metric $ds^2 = a^2(\eta)(d\eta^2 - \delta_{ij}dx^i dx^j)$, signature $(+,-,-,-)$:

$$g_{\mu\nu} = a^2\,\mathrm{diag}(1,-1,-1,-1), \quad g^{\mu\nu} = a^{-2}\,\mathrm{diag}(1,-1,-1,-1), \quad \sqrt{-g} = a^4. \tag{A.1}$$

The non-vanishing Christoffel symbols are

$$\Gamma^0_{00} = \mathcal{H}, \quad \Gamma^0_{ij} = \mathcal{H}\,\delta_{ij}, \quad \Gamma^i_{0j} = \mathcal{H}\,\delta^i_j, \tag{A.2}$$

where $\mathcal{H} = a'/a$ is the conformal Hubble parameter.

A.2 Comoving Observer

The comoving observer has four-velocity

$$u^\mu = (a^{-1}, 0, 0, 0), \quad u_\mu = g_{\mu\nu}u^\nu = (a, 0, 0, 0), \quad u^\mu u_\mu = 1. \tag{A.3}$$

A.3 Electric and Magnetic Fields in Coulomb Gauge

In Coulomb gauge ($A_0 = 0$, $\partial_i A^i = 0$):

$$F_{0i} = \partial_\eta A_i = A_i', \quad F_{ij} = \partial_i A_j - \partial_j A_i. \tag{A.4}$$

Electric field:

$$E_i \equiv F_{i\nu}u^\nu = F_{i0}u^0 = -F_{0i}/a = -A_i'/a. \tag{A.5}$$

Magnetic field (spatial components):

$$B^i = \frac{1}{2}\varepsilon^{i0jk}u_0 F_{jk} = \frac{a}{2}\cdot\frac{\varepsilon^{ijk}}{a^4}\cdot F_{jk} = \frac{\varepsilon^{ijk}F_{jk}}{2a^3} = \frac{\varepsilon^{ijk}\partial_j A_k}{a^3}, \tag{A.6}$$

where we used $\varepsilon^{i0jk} = -\varepsilon^{0ijk}$ and $\varepsilon^{0ijk} = \varepsilon^{ijk}/a^4$ (Levi-Civita tensor).

Rescaled fields ($\mathcal{E}_i \equiv a^2 E_i$, $\mathcal{B}_i \equiv a^2 B_i$):

$$\mathcal{E}_i = a^2 \cdot (-A_i'/a) = -aA_i'$$

(Wait: the physical field is $E_i^{\mathrm{phys}} = -A_i'/a$, so $\mathcal{E}_i = a^2 E_i^{\mathrm{phys}} = -aA_i'$).

Hmm—let us be more careful. The paper defines the rescaled fields so that the equations take the Minkowski form. From equation (32) of the paper: $\mathcal{E}_i = -A'_i$ and $\mathcal{B}_i = -\varepsilon_{ijk}\partial_j A_k$. These follow from the rescaled definitions $\mathcal{E} \equiv a^2\boldsymbol{E}$ and $\mathcal{B} \equiv a^2\boldsymbol{B}$ where $\boldsymbol{E}$ and $\boldsymbol{B}$ are the *physical* 3-vector fields (not covariant components), which satisfy $E^i = -A'^i/a^2$ and $B^i = \varepsilon^{ijk}\partial_j A_k/a^2$ (from raising indices with $-a^{-2}\delta^{ij}$ and δ_{ij} respectively):

$$\mathcal{E}_i = a^2 E_i^{\rm phys} = a^2 \cdot \frac{-A'_i}{a^2} = -A'_i, \quad \mathcal{B}_i = a^2 B_i^{\rm phys} = a^2 \cdot \frac{-\varepsilon_{ijk}\partial_j A_k}{a^2} = -\varepsilon_{ijk}\partial_j A_k. \quad \text{(A.7)}$$

This confirms equation (3.7). □

A.4 Klein-Gordon and Modified Maxwell Equations

From the Lagrangian (2.1), the equations of motion are:

$$\frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}\,\partial^\mu\phi) + m^2\phi = \frac{1}{4}g_{\phi\gamma}F_{\mu\nu}\tilde{F}^{\mu\nu}, \quad \text{(A.8)}$$

$$\partial_\mu(\sqrt{-g}\,F^{\mu\nu}) = g_{\phi\gamma}\,\partial_\mu(\sqrt{-g}\,\phi\,\tilde{F}^{\mu\nu}). \quad \text{(A.9)}$$

In Coulomb gauge, writing out equation (A.9) for $\nu = i$ (spatial):

$$\partial_0(\sqrt{-g}\,F^{0i}) + \partial_j(\sqrt{-g}\,F^{ji}) = g_{\phi\gamma}\left[\partial_0(\sqrt{-g}\,\phi\,\tilde{F}^{0i}) + \partial_j(\sqrt{-g}\,\phi\,\tilde{F}^{ji})\right]. \quad \text{(A.10)}$$

Left-hand side:

$$F^{0i} = g^{00}g^{ij}F_{0j} = a^{-2}(-a^{-2}\delta^{ij})A'_j = -a^{-4}A'^i, \quad \text{(A.11)}$$

$$\sqrt{-g}\,F^{0i} = a^4 \cdot (-a^{-4}A'^i) = -A'^i. \quad \text{(A.12)}$$

$$F^{ji} = g^{jj'}g^{ii'}F_{j'i'} = (-a^{-2})^2\delta^{jj'}\delta^{ii'}F_{j'i'} = a^{-4}F^{ji}_{\rm flat}, \quad \text{(A.13)}$$

$$\sqrt{-g}\,F^{ji} = F^{ji}_{\rm flat} = \delta^{jj'}\delta^{ii'}(\partial_{j'}A_{i'} - \partial_{i'}A_{j'}) = \partial^j A^i - \partial^i A^j. \quad \text{(A.14)}$$

Therefore:

$$\text{LHS} = \partial_\eta(-A'^i) + \partial_j(\partial^j A^i - \partial^i A^j) = -A''^i + \nabla^2 A^i - \partial^i(\nabla\cdot\boldsymbol{A}) = -A''^i + \nabla^2 A^i, \quad \text{(A.15)}$$

where we used the Coulomb gauge condition $\nabla\cdot\boldsymbol{A} = 0$.

Right-hand side: Using $\sqrt{-g}\,\tilde{F}^{\mu\nu} = \frac{1}{2}\bar{\varepsilon}^{\mu\nu\alpha\beta}F_{\alpha\beta}$ (with $\bar{\varepsilon}^{0123} = +1$):

$$\sqrt{-g}\,\tilde{F}^{0i} = \frac{1}{2}\bar{\varepsilon}^{0ijk}F_{jk} = \frac{1}{2}\varepsilon^{ijk}F_{jk} = \varepsilon^{ijk}\partial_j A_k = -\mathcal{B}^i \quad \text{(cf. eq. (A.7))}, \quad \text{(A.16)}$$

$$\sqrt{-g}\,\tilde{F}^{ji} = \frac{1}{2}\bar{\varepsilon}^{ji0k}F_{0k} = \frac{1}{2}(-\bar{\varepsilon}^{ij0k})F_{0k} = -\frac{1}{2}\bar{\varepsilon}^{ij0k}A'_k. \quad \text{(A.17)}$$

For the background ($\bar{\phi}$, $\bar{F}$) we have $\mathcal{E} = 0$, so:

$$\text{RHS(background)} = g_{\phi\gamma}\partial_\eta(\phi\cdot(-\mathcal{B}^i)) = -g_{\phi\gamma}(\phi'\,\mathcal{B}^i + \phi\,\mathcal{B}'^i). \quad \text{(A.18)}$$

Linearising: We now decompose $\phi \to \bar{\phi} + \delta\phi$ and $A_\mu \to \bar{A}_\mu + \delta A_\mu$ (where δA_μ gives the secondary photon field, which we call A_μ for short). The background satisfies its own Maxwell equation. To linear order in the fluctuations:

$$\begin{aligned} -A''^i + \nabla^2 A^i &= g_{\phi\gamma}\partial_\eta\big(-(\delta\phi)\,\bar{\mathcal{B}}^i\big) + g_{\phi\gamma}\partial_j\left(-\frac{1}{2}\bar{\varepsilon}^{ij0k}(\delta\phi)\bar{A}'_k\right) \\ &= -g_{\phi\gamma}\left[(\delta\phi)'\bar{\mathcal{B}}^i + (\delta\phi)\bar{\mathcal{B}}'^i\right] - g_{\phi\gamma}\varepsilon^{ijk}\partial_j(\delta\phi)\bar{\mathcal{E}}_k. \end{aligned} \quad \text{(A.19)}$$

Setting $\bar{\mathcal{E}} = 0$ and noting that the second term $(\delta\phi)\bar{\mathcal{B}}'$ is suppressed by $\mathcal{H}/k \ll 1$ for sub-Hubble modes (the variation of $\mathcal{B}$ is slow compared to the mode oscillation), the dominant term is the first one, giving:

$$A_i'' - \nabla^2 A_i = -g_{\phi\gamma}\,(\delta\phi)'\,\mathcal{B}_i, \tag{A.20}$$

which is equation (2) of the paper with source (3) (dropping the electric field term). □

B The DCBH Scenario: Physical Background

B.1 Why Do We Need Big Seed Black Holes?

Eddington-limited accretion grows a black hole exponentially with an e-folding time (the Salpeter time) of

$$t_{\rm Sal} = \frac{M}{\dot{M}_{\rm Edd}} = \frac{\sigma_T c}{4\pi G m_p} \cdot \frac{\eta}{1-\eta} \simeq 45\,\eta_{0.1}\,{\rm Myr}, \tag{B.1}$$

where $\eta \simeq 0.1$ is the radiative efficiency and σ_T is the Thomson cross section. From $z = 6$ ($t \approx 900\,{\rm Myr}$) back to the epoch of seed formation ($z \sim 20$–30, $t \approx 100$–$200\,{\rm Myr}$), the available time is 700–800 Myr, corresponding to $\approx$ 15–18 Salpeter times, i.e. a mass amplification of e^{15}–$e^{18} \sim 10^6$–10^8. A seed of mass $\sim 100\,M_\odot$ would grow to only $\sim 10^8\,M_\odot$, which is already tight. To reach $10^{10}\,M_\odot$ within the observed timescale requires seeds of $\sim 10^4$–$10^6\,M_\odot$.

B.2 Molecular Hydrogen Cooling and the Lyman-Werner Band

Neutral hydrogen cools very inefficiently for $T < 10^4\,{\rm K}$. Molecular hydrogen (H_2) forms at $T \lesssim 10^4\,{\rm K}$ and is a far more efficient coolant, allowing gas to cool to $T \sim 200\,{\rm K}$. This enables Jeans fragmentation into solar-mass objects. The Lyman-Werner photons (11.2–13.6 eV) dissociate H_2 via the Solomon process:

$$\mathrm{H_2} + h\nu_{\rm LW} \to \mathrm{H_2^*} \to \mathrm{H} + \mathrm{H}. \tag{B.2}$$

A sufficient flux of LW photons therefore maintains the gas at $T > 10^4\,{\rm K}$, preventing fragmentation and allowing monolithic collapse into a DCBH.

B.3 Critical LW Flux

Numerical simulations of gas collapse in the presence of an LW background find a critical flux

$$J_{\rm LW,crit} \sim 10^2\text{–}10^3\,J_{21}, \quad J_{21} \equiv 10^{-21}\,{\rm erg\,s^{-1}cm^{-2}Hz^{-1}sr^{-1}}, \tag{B.3}$$

which corresponds in natural units to

$$\Phi_{\rm LW,crit} \sim 10^{-44}\text{–}10^{-43}\,{\rm GeV}^3. \tag{B.4}$$

C Conversion Factors and Key Formulae

C.1 Conversion Factors

$$
\begin{aligned}
1\,\mathrm{G} &= 1.95\times 10^{-2}\,\mathrm{eV}^2 = 1.95\times 10^{-20}\,\mathrm{GeV}^2, &\text{(C.1)}\\
1\,\mathrm{eV} &= 1.16\times 10^{4}\,\mathrm{K}\ \ (k_B T), &\text{(C.2)}\\
M_{\mathrm{pl}} &= (8\pi G)^{-1/2} = 2.435\times 10^{18}\,\mathrm{GeV} = 2.435\times 10^{27}\,\mathrm{eV}, &\text{(C.3)}\\
T_{\mathrm{rec}} &\simeq 0.26\,\mathrm{eV} \simeq 3000\,\mathrm{K}, &\text{(C.4)}\\
a_{\mathrm{rec}} &= 1/1100, \quad H_{\mathrm{rec}} \simeq 10^{-29}\,\mathrm{eV}, &\text{(C.5)}\\
k_1 &\equiv \frac{2\pi}{1\,\mathrm{Mpc}} \simeq 10^{-29}\,\mathrm{eV}, &\text{(C.6)}\\
k_{\mathrm{LW}} &\sim 10\,\mathrm{eV}\ (11.2\text{–}13.6\,\mathrm{eV\ photon\ energy}). &\text{(C.7)}
\end{aligned}
$$

C.2 Summary of Key Equations

$$
\begin{aligned}
\text{ALP Lagrangian:}&\quad \mathcal{L} \supset \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - g_{\phi\gamma}\phi F\tilde{F}, &\text{(C.8)}\\
\text{DM oscillation:}&\quad \phi(\eta) = \phi_0\cos(m\eta), \quad \phi_0 \sim T_{\mathrm{rec}}^2/m, &\text{(C.9)}\\
\text{Critical wavenumber:}&\quad k_{\mathrm{c}} \simeq g_{\phi\gamma} m\phi_0\, a(\eta), &\text{(C.10)}\\
\text{DM fluctuation:}&\quad \delta\phi(k) = \tfrac{1}{2}\phi_0\,\mathcal{A}^{1/2}\,k^{-3/2}, \quad \mathcal{A}\simeq 10^{-9}, &\text{(C.11)}\\
\text{Photon flux:}&\quad \Phi_A(k) \sim g_{\phi\gamma}^2 B^2\,\mathcal{A}\,\phi_0^2 k^{-1}, &\text{(C.12)}\\
\text{At LW band:}&\quad \Phi_A(10\,\mathrm{eV}) \sim \tilde{g}_{\phi\gamma}^2 m_{20}^{-2}\times 10^{-39}\,\mathrm{GeV}^3, &\text{(C.13)}\\
\text{DCBH lower bound:}&\quad \tilde{g}_{\phi\gamma}^2/m_{20}^2 \gtrsim 10^{-5}, &\text{(C.14)}\\
\text{ARCADE2 upper bound:}&\quad \tilde{g}_{\phi\gamma}^2/m_{20}^2 \lesssim 4\times 10^{5}, &\text{(C.15)}\\
\text{EDGES upper bound:}&\quad \tilde{g}_{\phi\gamma}^2/m_{20}^2 \lesssim 5\times 10^{4}. &\text{(C.16)}
\end{aligned}
$$

Bibliography

[1] R. Brandenberger, J. Fröhlich and H. Jiao, "Cosmological Magnetic Fields from Ultralight Dark Matter," *Phys. Rev. Lett.* **136**, 031001 (2026) [arXiv:2502.19310 [hep-ph]].

[2] V. Kamali and R. Brandenberger, "Late Time Magnetogenesis from Ultralight Scalar Dark Matter," [arXiv:2601.00443 [astro-ph.CO]] (2026).

[3] H. Jiao, R. Brandenberger and V. Kamali, "Direct-collapse supermassive black holes from ultralight dark matter," *Phys. Rev. D* **112**, 043546 (2025) [arXiv:2503.19414 [astro-ph.CO]].

[4] N. Brahma and R. Brandenberger, "Parametric Resonance and Backreaction Effects in Magnetogenesis from Ultralight Dark Matter," [arXiv:2602.04285 [astro-ph.CO]] (2026).

[5] A. M. Taylor, I. Vovk and A. Neronov, "Extragalactic magnetic fields constraints from simultaneous GeV-TeV observations of blazars," *Astron. Astrophys.* **529**, A144 (2011).

[6] R. Durrer and A. Neronov, "Cosmological Magnetic Fields: Their Generation, Evolution and Observation," *Astron. Astrophys. Rev.* **21**, 62 (2013).

[7] N. Dalal and A. Kravtsov, "Excluding fuzzy dark matter with sizes and stellar kinematics of ultrafaint dwarf galaxies," *Phys. Rev. D* **106**, 063517 (2022).

[8] D. J. Fixsen *et al.*, "ARCADE 2 Measurement of the Extra-Galactic Sky Temperature at 3-90 GHz," *Astrophys. J.* **734**, 5 (2011).

[9] J. D. Bowman *et al.*, "An absorption profile centred at 78 megahertz in the sky-averaged spectrum," *Nature* **555**, 67 (2018).

[10] J. H. Traschen and R. H. Brandenberger, "Particle Production During Out-of-equilibrium Phase Transitions," *Phys. Rev. D* **42**, 2491 (1990).

[11] A. Kushwaha and R. Brandenberger, "ALP Dark Matter, Cosmological Magnetic Fields and the Direct Collapse Black Hole Formation Scenario," [arXiv:2602.22170 [hep-ph]] (2026). *[Primary source for these notes.]*